

RAFFLES INSTITUTION
2018 Year 6 Preliminary Examination

Higher 2

CANDIDATE
NAME

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CIVICS
GROUP

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NUMBER

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BIOLOGY

9744/03

Paper 3 Long Structured and Free-response Questions

18th September 2018

2 hours

Candidates answer on the Question Paper.
Additional Materials: Writing paper.

READ THESE INSTRUCTIONS FIRST

Write your index number, CT group & name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use a HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

DO NOT WRITE IN ANY BARCODES

Section A

Answer **all** questions in the spaces provided on the Question Paper.

Section B

Answer any **one** question in the writing paper provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, **hand in your essay question SEPARATELY.**

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
Section A	
1	/ 27
2	/ 23
Section B	
3 or 4	/ 25
Total	/ 75

This document consists of **14** printed pages.



Raffles Institution
Internal Examination

Section A

Answer **all** the questions in this section.

- 1** Lactose is the major carbohydrate in milk which is a main source of food for infants. Young children almost universally produce lactase, an enzyme which digests lactose.

Fig. 1.1 shows the structure of a lactose molecule.

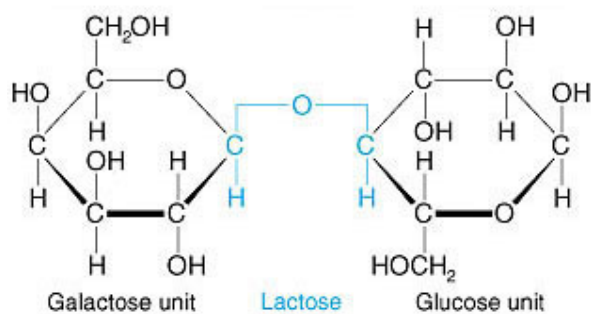


Fig. 1.1

- (a)** Show how a lactose molecule is hydrolysed into its monomers in the space provided below.

[2]

(b) The structure of lactase is shown in Fig. 1.2.

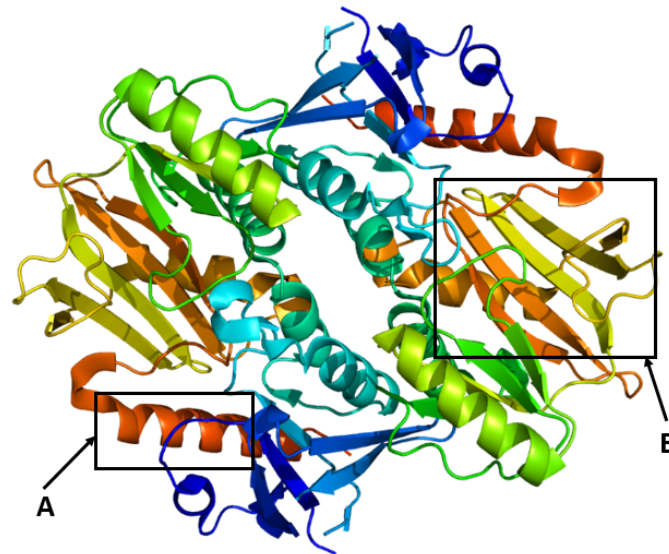


Fig. 1.2

Compare the structures labelled **A** and **B** in Fig. 1.2.

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..... [3]

- (c) Lactose is digested in humans by lactase in the small intestine, which has an alkaline environment.

People who are lactose intolerant lack the enzyme lactase, which digests lactose. If undigested, lactose reaches the large intestine where it can be fermented by microbes. Lactose fermentation results in the production of volatile fatty acids and gases such as carbon dioxide and hydrogen gas. A part of the hydrogen gas is expired by breath. The hydrogen (H₂) breath test has been developed as a method to identify lactose intolerance.

The fungus, *Aspergillus oryzae*, produces a lactase which converts lactose to glucose and galactose at a pH of 2 to 3.5. This makes the fungal lactase particularly effective to digest lactose in the stomach, which has a highly acidic environment. Therefore it is ideal for use in dietary supplements as a digestive aid.

- (i) Explain a similarity and a difference in structure between the fungus lactase and the human lactase.

Similarity.....

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Difference.....

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..... [4]

- (ii) Explain why the fungal lactase will only work in the stomach but not in the small intestine.

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- (d) To determine if the fungal lactase can reduce the symptoms of lactose intolerance, 100 lactose-intolerant participants took part in a H_2 breath test.

All the participants were given 500 mL of milk during the test. Following that, they were divided into 3 groups: 1 group was given a placebo (0 ALU of lactase). The other 2 groups were given low dosage (4625 ALU) and high dosage (7700 ALU) of lactase, respectively.

The graph in Fig. 1.3 shows the results of the H_2 breath test.

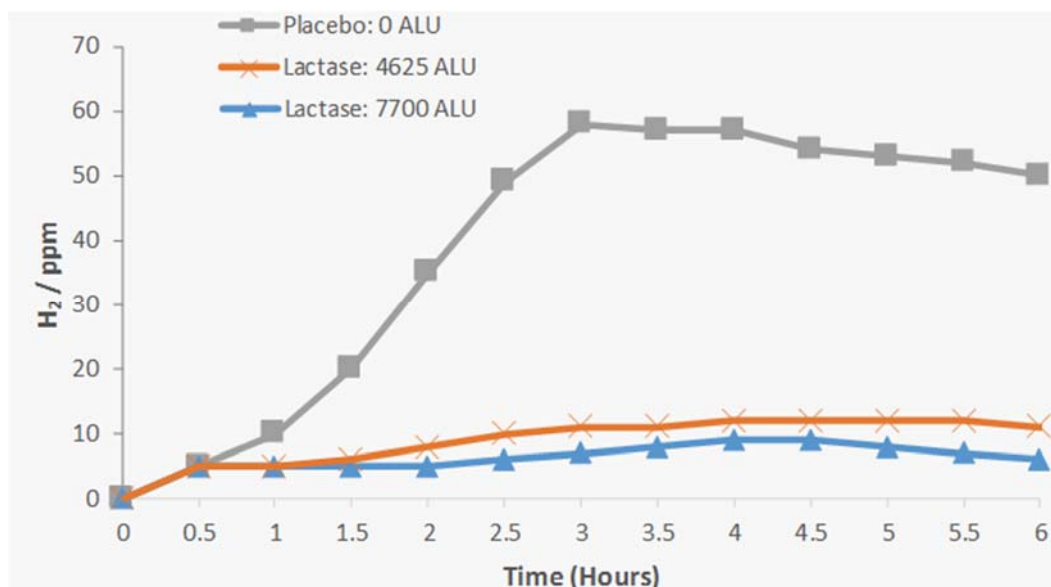


Fig. 1.3

- (i) With reference to Fig. 1.3, discuss the effectiveness of lactase treatment in reducing the symptoms of lactose intolerance.

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..... [3]

- (ii) Explain the shape of the graph for the group given placebo.

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..... [2]

- (iii) The participants are required to fast for 12 hours from milk-containing food as a prerequisite for doing the H_2 breath test.

A participant who did not fast from milk-containing food is given the low dosage (4625 ALU) of lactase. He drank milk 4 hours before the H_2 breath test.

Draw on the graph in Fig. 1.3, the results obtained from this participant's test. [1]

- (e) In most mammals, including 65% of humans, the level of lactase mRNA found in the intestinal epithelial cell is greatly decreased in adults.

Describe a mechanism that accounts for this age-dependent lactase regulation during adulthood.

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..... [2]

- (f) Some individuals, however, are lactase-persistent (LP). They maintain high levels of lactase expression throughout adult life and hence remain tolerant of lactose. LP is thought to have originated from a mutation.

The pedigree in Fig. 1.4 shows the presence of lactase persistence of a family.

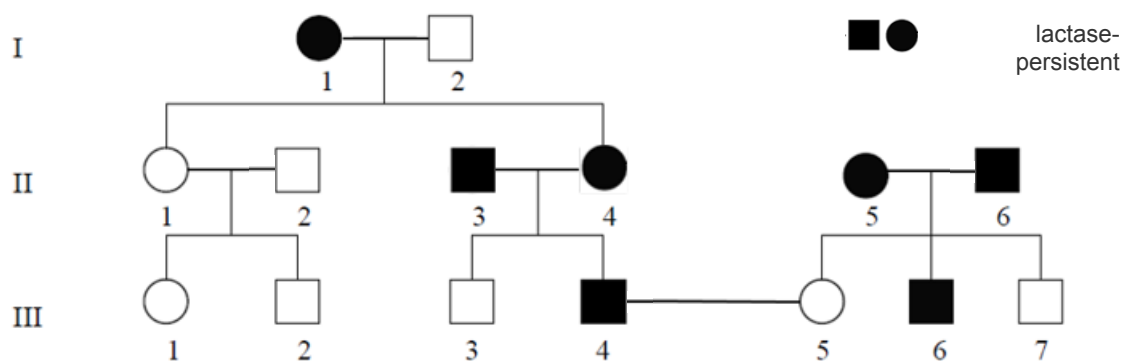


Fig. 1.4

- (i) State the mode of inheritance of lactase persistence.

..... [1]

(ii) Explain how you arrived at your answer in (f)(i).

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..... [4]

- (g) A single nucleotide polymorphism (SNP) is found to exist between LP and non-LP individuals. The SNP known as LCT₋₁₃₉₁₀ has 2 alleles, T allele and C allele. The genotype of individuals at that SNP can be used to predict if they are lactase-persistent.

The T allele is found in LP individuals who have higher levels of lactase. Individuals who carry the C allele have lower levels of lactase.

In a RFLP assay of both the genetic variants, polymerase chain reaction was used to amplify both the T and C alleles. Each allele resulted in a 448 bp product. The amplified products were digested with restriction enzyme *BsmFI*. It produced two fragments (351 and 97 bp) for the C allele, and three fragments (253, 98 and 97 bp) for the T allele.

Fig. 1.5 shows a gel with the results of the RFLP analysis for an individual who is homozygous for the T allele. Draw on Fig. 1.5 to show the results of the banding patterns of individuals II 5 and III 1 from Fig. 1.4.

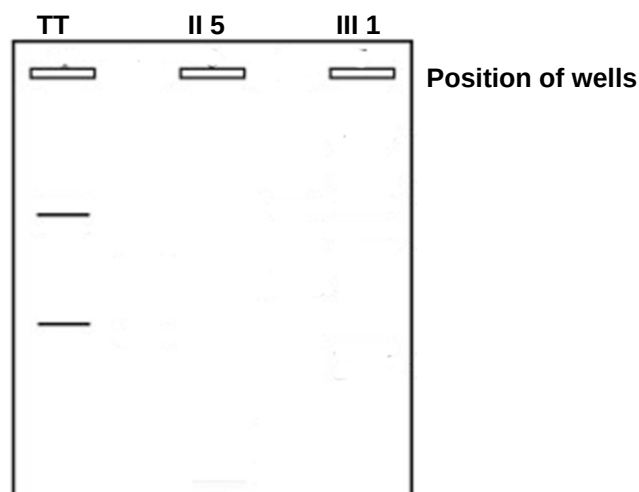


Fig. 1.5

[2]

[Total: 27]

- 2 Dengue fever is a mosquito-borne disease caused by the dengue virus. Dengue is transmitted by several species of mosquito, including *Aedes aegypti*. Symptoms such as high fever, headache, vomiting, muscle and joint pains, and a characteristic skin rash, typically begin three to fourteen days after infection.

The disease can develop into the life-threatening dengue hemorrhagic fever.

A number of tests are available to confirm the diagnosis of dengue fever. This includes the Enzyme Linked Immunosorbant Assay (ELISA) which detects the presence of a patient's immunoglobulin (Ig) produced against the dengue virus' non-structural-1 protein (NS-1). This process takes about 3 hours.

Fig. 2.1 shows the principle of ELISA.

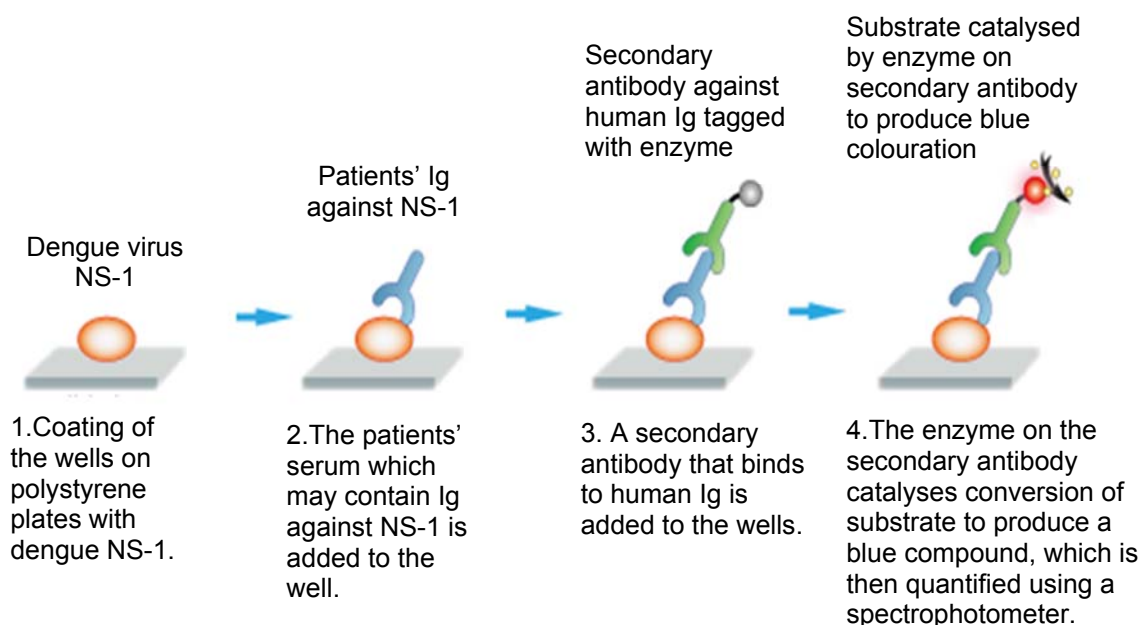


Fig. 2.1

- (a) Explain how the structure of the patient's Ig enables its binding to the NS-1 that coats the wells of the polystyrene plates.

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..... [2]

- (b) ELISA allows detection for the presence of patients' Ig. It also allows the quantification of the concentration of human Ig against NS-1.

With reference to Fig. 2.1 and using your knowledge of enzymes, explain how the process of ELISA allows quantification of the concentration of human Ig against NS-1.

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..... [3]

- (c) An alternative test is the quantitative polymerase chain reaction (qPCR) that detects the presence of dengue viral RNA. The principles of qPCR is similar to normal PCR except that it uses RNA as a template. In qPCR, SYBR® Green is added to the reaction mix. Interaction between SYBR® Green with the PCR products produces green fluorescence, which can be detected.

Fig. 2.2 shows the primer annealing and extension phases of qPCR.

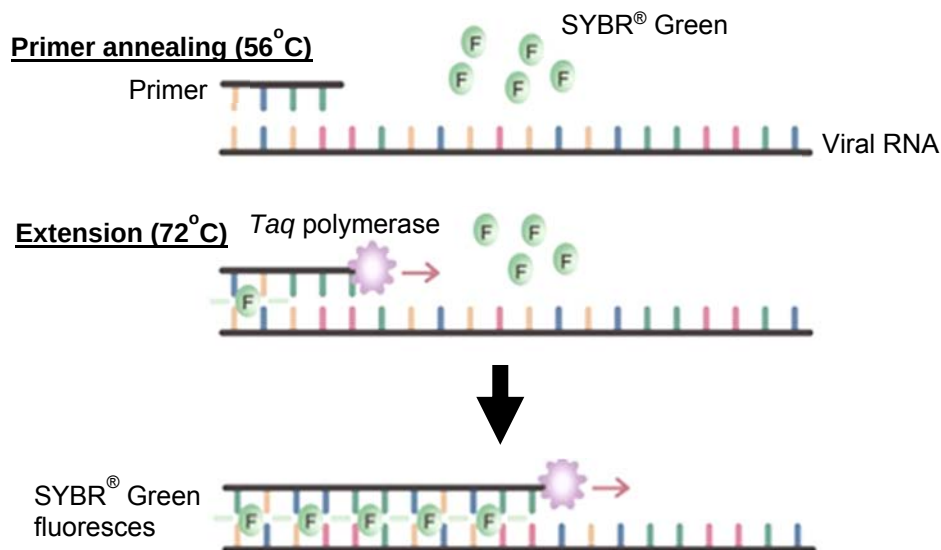


Fig. 2.2

- (i) Describe one characteristic of primers used and how it facilitates qPCR.

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..... [2]

- (ii) Suggest how SYBR® green interacts with the PCR products.

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 [1]

- (d) qPCR is performed in a thermocycler that measures the fluorescence in the tube over a number of cycles. If the level of fluorescence is above the threshold level (4 A.U.), it is considered a positive test.

Fig. 2.3 shows the changes in the fluorescence level over 30 cycles.

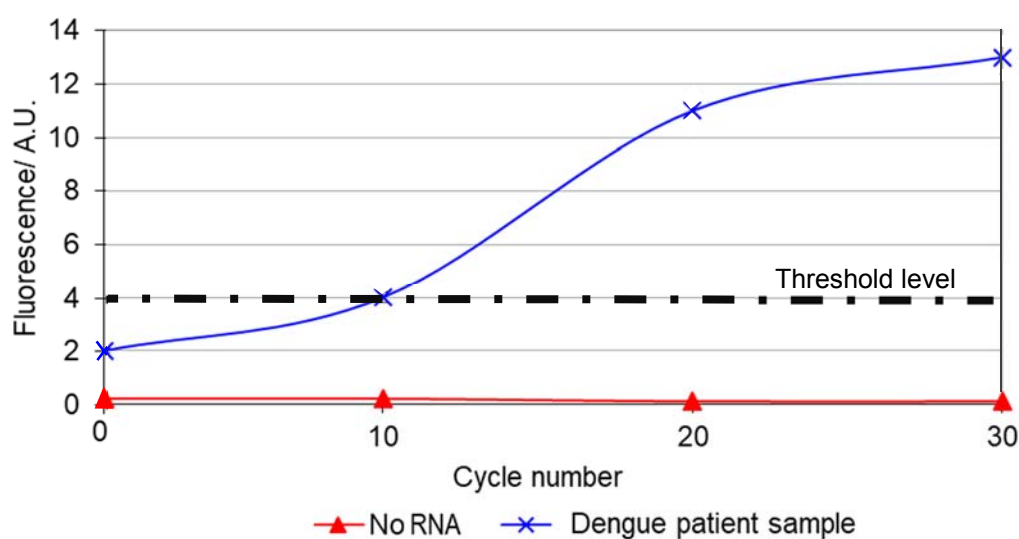


Fig. 2.3

- (i) Explain the purpose of including the control with no RNA.

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 [1]

- (ii) If a typical qPCR cycle takes 3 minutes, calculate the minimum time taken to report a positive result.

Minimum time taken is

[2]

(iii) Suggest why qPCR is a better test compared to ELISA.

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..... [2]

(e) Factors affecting the transmission of dengue are being closely monitored.

Singapore is not insulated from the effects of global warming. Fig. 2.4 shows the annual average temperature in the past 70 years.

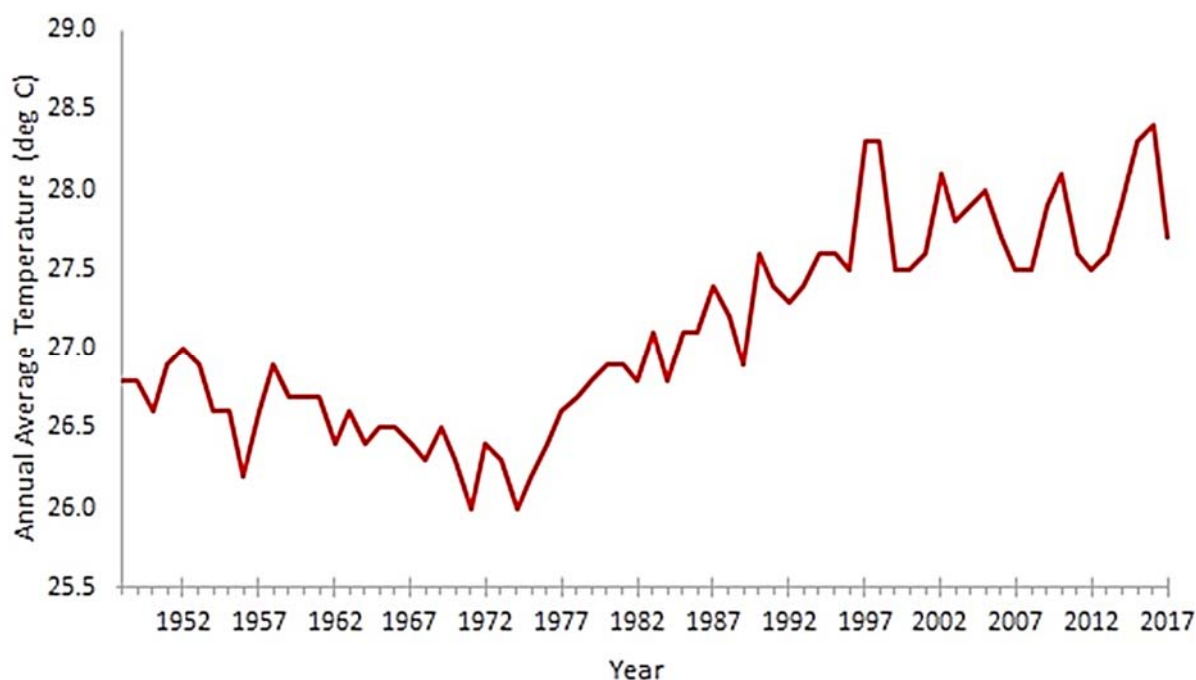


Fig. 2.4

With reference to Fig. 2.4,

(i) describe and explain the temperature trends in Singapore from 1974 to 2017,

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..... [2]

- (ii) explain the impact of climate change on transmission of dengue in Singapore.

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As it is situated near the equator, Singapore's climate is characterised by high temperatures, rainfall and humidity all year-round. However, there exists slight variations within the months.

The figures below show the variations in temperature (Fig. 2.4a) and rainfall (Fig. 2.4b) over the months in the year of 2015.

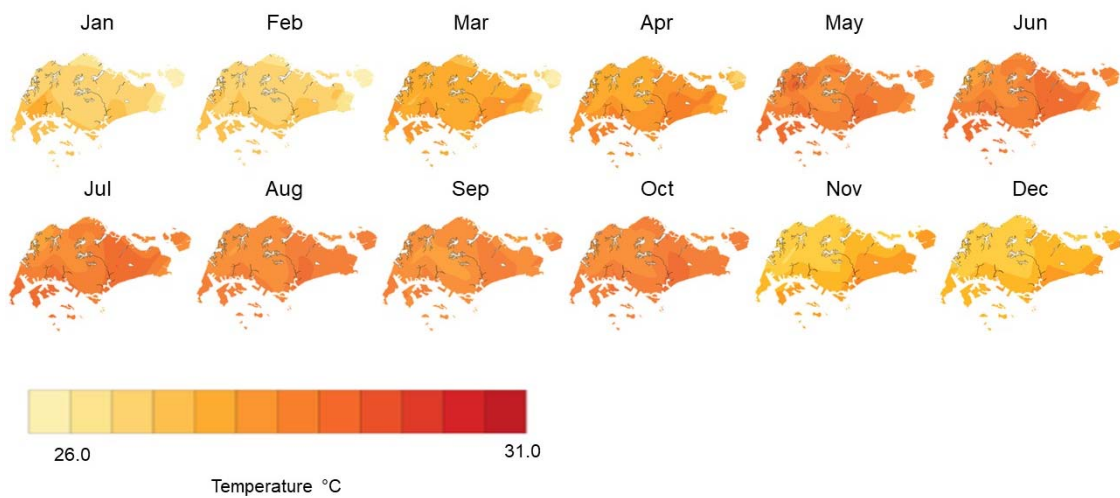


Fig. 2.4a

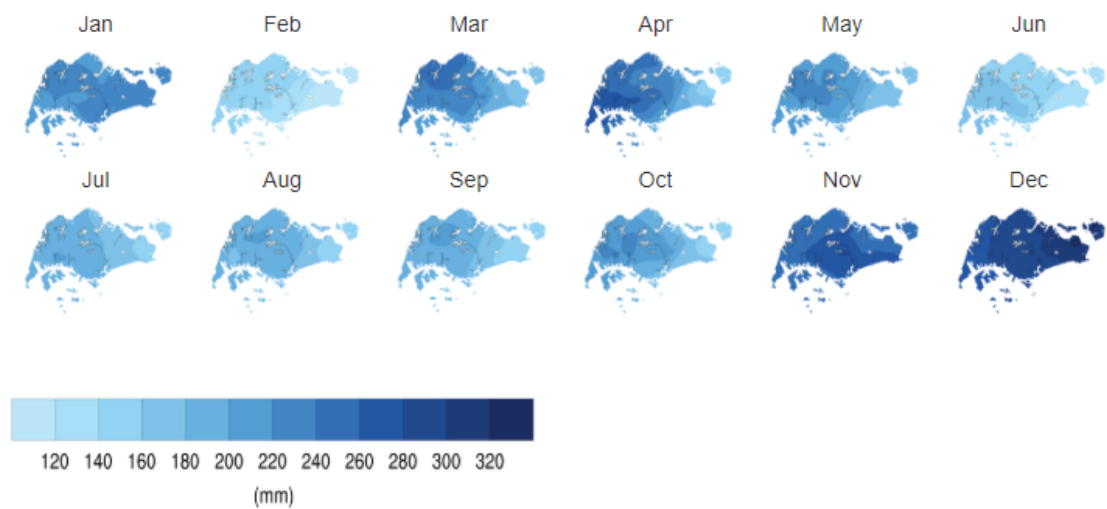


Fig. 2.4b

- (iii) It has been observed that there are higher incidences of dengue infections reported between June to October.

With reference to Fig. 2.4a and 2.4b, and the information given above, comment on the aspect of climate which has a greater influence on dengue transmission in Singapore.

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 [2]

- (f) In order to decrease dengue transmission, studies have been conducted to find a suitable method of control of the *Aedes* mosquito population.

One such method implemented recently was the use of *Wolbachia*, a bacteria common in insect species, to control the mosquito population. *Wolbachia* infection in mosquitoes are maternally transferred, and mosquitoes carrying the *Wolbachia* bacteria are unable to transmit the dengue virus.

Fig. 2.5 shows the crosses of *Wolbachia*-infected mosquitoes.

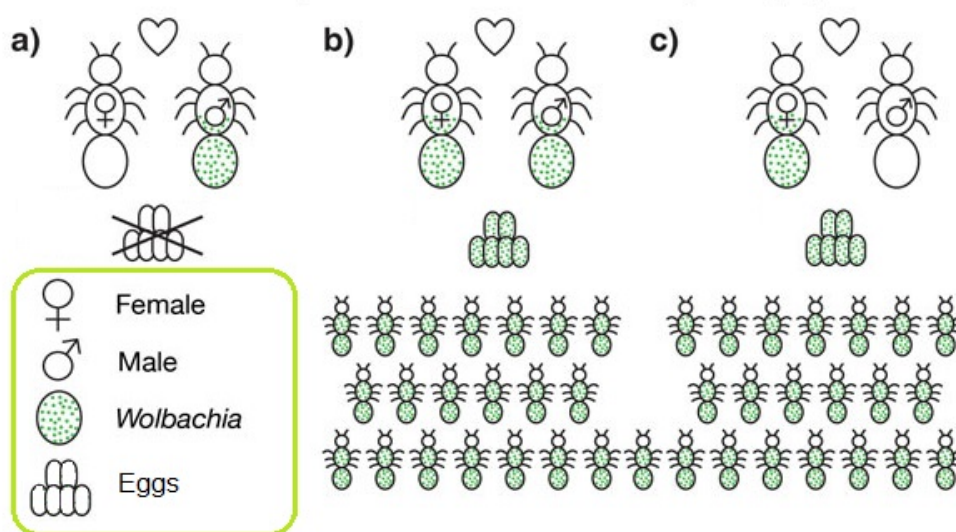


Fig. 2.5

When a male *Wolbachia*-infected mosquito mates with a wild-type female mosquito, the female mosquito lays eggs which are unable to hatch, due to a phenomenon known as cytoplasmic incompatibility. A scientist hypothesised the following to explain cytoplasmic incompatibility: after fertilisation, condensation of paternal chromosomes in the zygote is prevented.

- (i) Given the above information, explain how nuclear division may be affected.

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..... [2]

- (ii) Different *Wolbachia* strategies have been implemented to control the spread of mosquito-borne diseases. Some countries have opted for the release of both *Wolbachia*-infected males and female mosquitoes. However, Singapore has adopted the strategy to release only *Wolbachia*-infected male mosquitoes.

Given the above information, justify Singapore's choice to adopt the release of only *Wolbachia*-infected males.

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..... [2]

[Total: 23]

Section B

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 3 (a) Account for the importance of hydrogen bonds in biological organisms. [13]

- (b) Discuss the diverse roles of membranes in a cell. [12]

[Total: 25]

- 4 (a) Discuss how the complementary shapes between biomolecules enable eukaryotic cells to carry out their diverse functions. [13]

- (b) Organisms are able to detect and respond to changes in their environments. Discuss this, with reference to specific examples in unicellular and multi-cellular organisms. [12]

[Total: 25]

End of Paper